JURNAL PRAKTIKUM PENGUKURAN DAN ALAT UKUR

AMPEREMETER

OLEH

231344067 DHANI IBRAHIM PRATAMA

231344080 MUTHIA APRILIANA AZIZAH

231344073 MUHAMMAD RAHMAN KURNIA

Kelompok 4

Tanggal Praktik : Senin, 25 September 2023

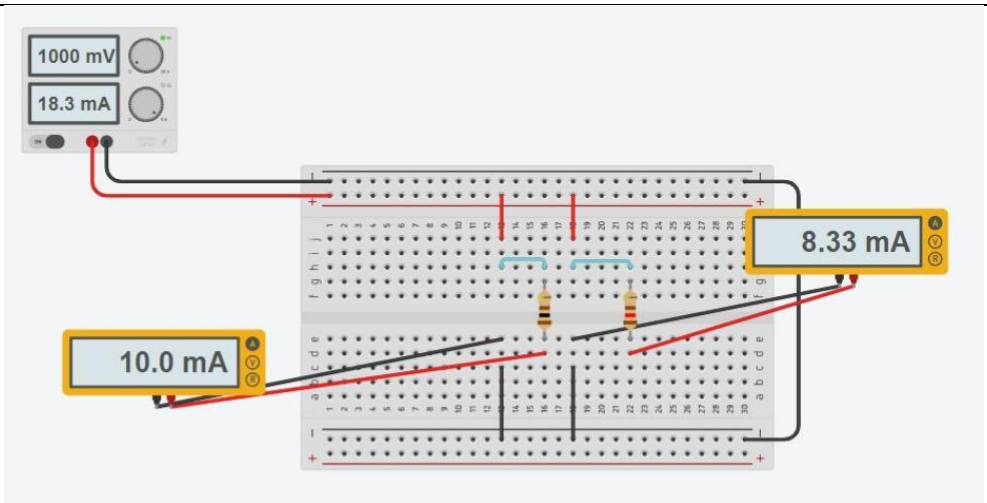
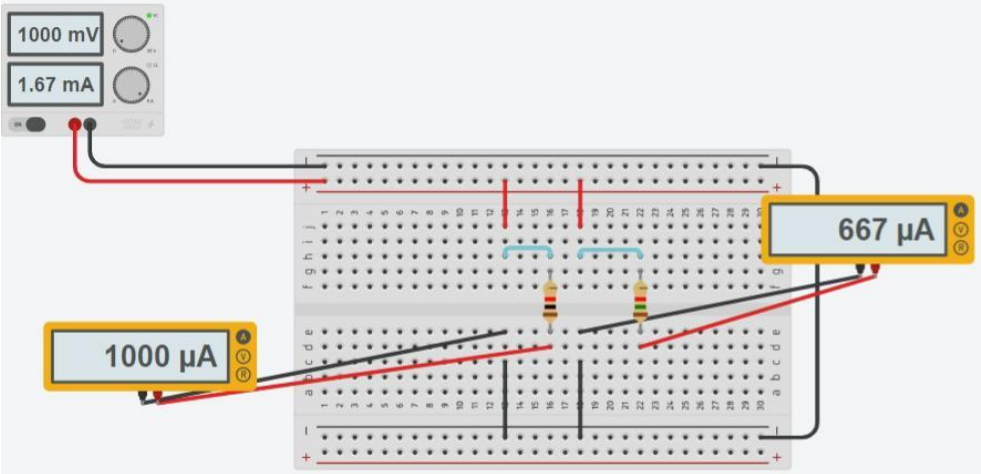
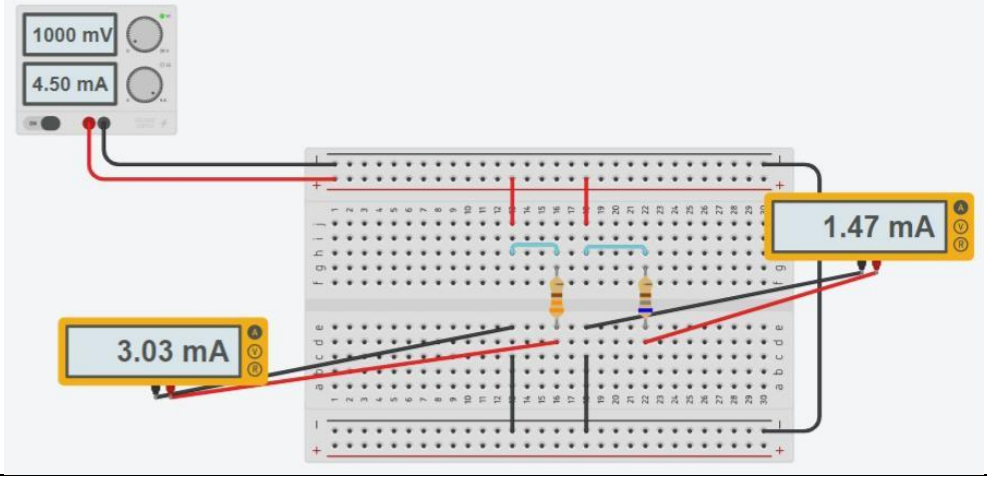
PROGRAM STUDI D4 TEKNIK TELEKOMUNIKASI

JURUSAN TEKNIK ELEKTRO

POLITEKNIK NEGERI BANDUNG

2023

1. Percobaan Pada Tinkercad

R (Ω)	Hasil Percobaan
R1 = 100 R2 = 120	
R1 = 1k R2 = 1,5k	
R1 = 330 R2 = 680	

2. Pengukuran Secara Teori

R1 = 100Ω R2 = 120Ω
$R_{Total} = R1//R2 = \frac{100 \times 120}{100 + 120} = 54{,}54\ \Omega$
$I = \frac{V}{R_{Total}} = \frac{1}{54{,}54} = 0{,}0183\ A$
$I_1 = \frac{R2}{R2 + R1} \times I = \frac{120}{120 + 100} \times 0{,}0183 = 0{,}00998\ A \approx 9{,}98\ mA$
$I_2 = \frac{R1}{R2 + R1} \times I = \frac{100}{120 + 100} \times 0{,}0183 = 0{,}00831\ A \approx 8{,}31\ mA$

R1 = 1kΩ R2 = 1,5kΩ
$R_{Total} = R1//R2 = \frac{1000 \times 1500}{1000 + 1500} = 600\ \Omega$
$I = \frac{V}{R_{Total}} = \frac{1}{600} = 0{,}00166\ A$
$I_1 = \frac{R2}{R2 + R1} \times I = \frac{1500}{1500 + 1000} \times 0{,}00166 = 0{,}000996 \approx 0{,}996\ mA \approx 996\ \mu A$
$I_2 = \frac{R2}{R2 + R1} \times I = \frac{1000}{1500 + 1000} \times 0{,}00166 = 0{,}000664 \approx 0{,}664\ mA\ \approx 664\ \mu A$

R1 = 330Ω R2 = 680Ω
$R_{Total} = R1//R2 = \frac{330 \times 680}{330 + 680} = 222{,}1782\ \Omega$
$I = \frac{V}{R_{Total}} = \frac{1}{222{,}1782} = 0{,}0045\ A$
$I_1 = \frac{R2}{R2 + R1} \times I = \frac{680}{680 + 330} \times 0{,}0045 = 0{,}00302\ A \approx 3{,}02\ mA$
$I_2 = \frac{R2}{R2 + R1} \times I = \frac{330}{680 + 330} \times 0{,}0045 = 0{,}00147\ A \approx 1{,}47\ mA$